Revision 2021 • Program Core • Drawing

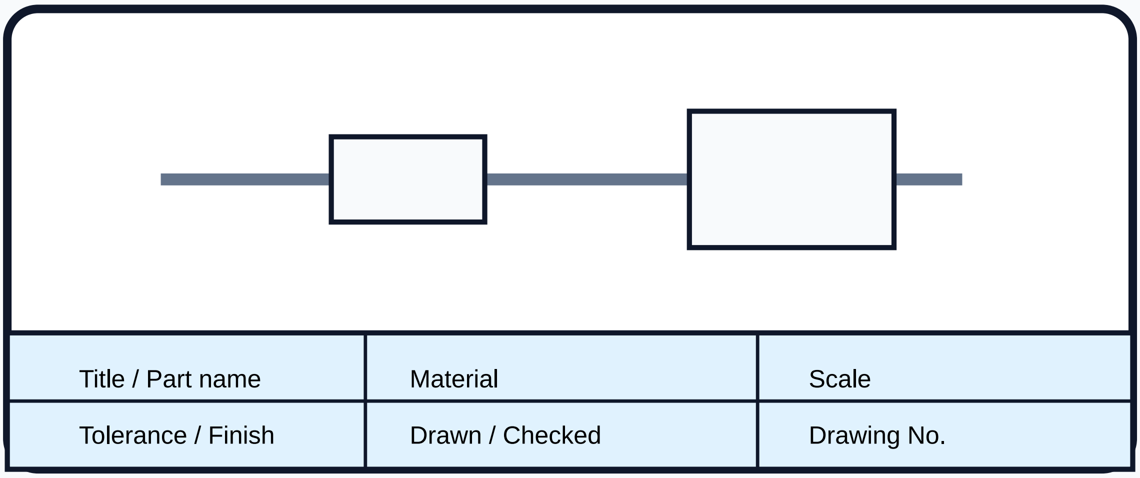
Machine Drawing

Professional diploma engineering handbook for course 3025. Built from the official syllabus for Mechanical Engineering / Manufacturing Technology. It covers fasteners, rivets, welds, limits, fits, tolerances, surface finish, assembly drawing and production drawing.

45
practical periods

1.5
credits

A2
drawing sheet focus



Production drawing = views + dimensions + tolerance + finish + title block

Course Overview

Machine Drawing is the communication link between design office, workshop, inspection and maintenance.

Official information

Item	Value

Course code	3025
Title	Machine Drawing
Program	Diploma in Mechanical Engineering / Manufacturing Technology
Semester	3 / 3
Credits	1.5
Category	Program Core
Periods	3 per week, 45 per semester

Prerequisite

Basic knowledge about all aspects of drawing and orthographic projection from Engineering Graphics.

> Drawing subject aanu. Reading only pora. Center line, view alignment, hatching, dimensions, BOM, title block ellam practice cheyyanam.

Course outcomes

CO	Outcome	Hours	Meaning
CO1	Outline fastening devices and assembly	7	Draw bolts, nuts, rivets, welds and locking arrangements.
CO2	Illustrate tolerances and surface finish	6	Use BIS tables, limits, fits, roughness and production notes.
CO3	Develop machine element drawings	19	Prepare assembled views, dimensions and BOM.
CO4	Develop production drawings	13	Prepare shop-ready part drawings with title block and inspection data.

Redesigned Syllabus Roadmap

1. Fasteners

Draw bolt-nut-washer, locking nuts, rivet heads, lap/butt joints and weld symbols. Exam importance: very high because neat standard proportions carry marks.

2. Fits and Finish

Explain limits, fits, tolerance zones, BIS table use, surface roughness values, grades, symbols and direction of lay.

3. Assemblies

Prepare A2 views and BOM for socket and spigot joint, knuckle joint, stuffing box, plummer block, flanged coupling and machine vice.

4. Production

Prepare production drawings of slip bush, stepped shaft, slotted nut and sleeve with dimensions, material, tolerance and surface finish.

Module 1: Fastening Devices and Assemblies

Textbook explanation

Threaded fasteners are temporary joints. Bolts pass through clearance holes and tighten with nuts; screws engage internal thread; studs have threaded ends. Machine drawing uses conventional thread representation instead of drawing actual helix.

Locking arrangements prevent loosening due to vibration. Common cases are lock nut, castle nut with split pin and spring washer. Riveted joints are permanent joints; lap and butt joints must show plates, straps, rivets, pitch and margin clearly.

Weld symbols communicate weld type, side, size and length using arrow line and reference line.

> Common mistakes: missing center line, wrong nut proportion, sectioning standard fasteners wrongly, confusing chain/zigzag rivets, placing weld symbol on wrong side.

What to draw

- Bolt, nut and plain washer assembly
- Nut with lock nut
- Castle nut with split pin
- Nut with spring washer
- Single/double riveted lap joints
- Single strap and double strap butt joints
- Weld symbols

Module 2: Limits, Fits, Tolerances and Surface Finish

Core explanation

A basic size is theoretical. Actual size is allowed between upper and lower limits. Tolerance is the permissible variation. Fit is the relationship between mating hole and shaft before assembly.

Clearance fit always gives looseness. **Interference fit** always gives tightness. **Transition fit** may give small clearance or small interference. BIS tables standardize tolerance zones and limit values.

Surface roughness describes fine irregularities left by manufacturing. Symbols indicate whether material removal is required, roughness value, lay and process note.

Formula examples

- Tolerance = upper limit - lower limit
- Minimum clearance = hole minimum - shaft maximum
- Maximum clearance = hole maximum - shaft minimum
- Positive clearance in all cases = clearance fit

Module 3: Assembly and Detail Drawings

This is the highest-duration module. Students must convert isometric/detail information into orthographic assembled views with sections, dimensions and bill of materials.

Required assemblies

- Socket and Spigot Joint
- Knuckle Joint
- Stuffing Box
- Plummer Block
- Flanged Coupling - unprotected type
- Machine Vice

Drawing procedure

1. Study every part and function.
2. Select elevation, plan and side view.
3. Draw center lines and main outline.
4. Use sectional view where internal relation is needed.

5. Add dimensions, balloons and BOM.
6. Check alignment, hatching and hidden lines.

Module 4: Production Drawings

Production drawing meaning

A production drawing is a complete manufacturing document for one component. It must include views, dimensions, tolerance, surface finish, material, scale, title block, drawing number, revision and notes.

Required syllabus components are slip bush, stepped shaft, slotted nut and sleeve. Use diameter symbols, center lines, chamfers, threads, tolerances and finish symbols properly.

Shop-floor checklist

- Material specified
- All functional dimensions given
- No duplicate/conflicting dimensions
- Tolerance/fits given where needed
- Surface finish marked on functional faces
- Title block complete

Formula / Rule Bank

1. Tolerance = upper limit - lower limit. Example: 40.000 - 39.975 = 0.025 mm.
2. Clearance fit: hole minimum greater than shaft maximum.
3. Interference fit: shaft minimum greater than hole maximum.
4. BOM rule: item number + part name + material + quantity.
5. Section rule: hatch cut material; do not section bolts/nuts longitudinally unless needed.
6. Production drawing rule: no material + no tolerance + no finish = not production-ready.

Practice and Expected Questions

Expected drawing questions

1. Draw bolt, nut and washer assembly.
2. Draw nut locking arrangements.
3. Draw chain and zigzag riveted lap joints.
4. Prepare plummer block assembly with BOM.
5. Prepare production drawing of stepped shaft.

Objective practice

1. Production drawing gives all manufacturing data. Answer: True.
2. BOM means Bill of Materials.
3. Castle nut is locked using split pin.
4. Clearance fit gives positive clearance.
5. Surface roughness is represented by texture symbols and values.

Fresh Model Question Paper

Part A - Short answers

1. Define tolerance.
2. List threaded fasteners.
3. State purpose of spring washer.
4. Define BOM.
5. State use of BIS tables.

Part B - Drawing / explanation

1. Draw bolt, nut and washer assembly.
2. Explain nut locking arrangements.
3. Draw riveted lap joints.
4. Explain limits, fits and tolerance zone.
5. Explain surface texture symbols.

Part C - Descriptive drawing

1. Prepare sectional views of knuckle joint with BOM.
2. Prepare stuffing box assembly.
3. Prepare production drawing of stepped shaft.
4. Prepare production drawing of slotted nut.

Complete Model Answers

Tolerance

Tolerance is permissible variation in a dimension. Formula: upper limit - lower limit. It supports interchangeability and inspection.

Bolt assembly

Draw common center line, bolt head, shank, washer and nut. Use standard proportions and conventional thread representation.

Assembly answer

Use correct section, hatching, item balloons, dimensions and bill of materials. Avoid unnecessary hidden lines.

Production drawing answer

Give views, dimensions, tolerance, surface finish, material, scale, drawing number, revision and title block.

References

P I Varghese and K C John; Sidheswar, Kannaiah and Sastry; N D Bhatt; P S Gill; V Lakshmi Narayan; M B Shah and B C Rana. Online resources: NDLI and NPTEL.